

# Shekinah Okunbo

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## EDUCATION

### Rice University

Expected Graduation: May 2029

*B.S. Electrical & Computer Engineering, AI Specialization; Minor in Data Science*

*Houston, TX*

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Embedded Systems, Digital Logic, Python Programming, Signals & Systems, NVIDIA Autonomous Robotics Course
- **Honors:** Rice Datathon 2026 — 1st Place (Underclassmen); NMC National Physics Olympiad 2025 — 4th nationally

## EXPERIENCE

### AI Agent Engineering Extern

Jul 2026 – Present

*Wayfair*

*Remote*

- Building multi-step AI workflows with n8n and Google Gemini to ingest product, blog, and social data, extract structured trend signals from unstructured text, and publish results to a live dashboard.

### Machine Learning Researcher

May 2026 – Jul 2026

*Rice Summer Undergraduate Research Fellowship*

*Houston, TX*

- Trained and evaluated five YOLO26 object-detection variants, building a reproducible pipeline for preprocessing, augmentation, training, and accuracy benchmarking on COCO and a custom desk-object dataset.
- Quantized and deployed GPU-trained models to an iPhone 16 Pro; profiled accuracy, latency, FPS, and model size across lighting and motion conditions to identify the best mobile speed-accuracy tradeoff.
- Achieved **81% mAP@50** and measured **30.08 FPS / 15.25 ms** for the fastest model versus 16.78 FPS / 48.53 ms for the largest, documenting reproducible deployment results.

### Software Developer

Sep 2025 – Present

*RiceApps*

*Rice University*

- Designed a normalized PostgreSQL/Supabase schema and SQL migrations; built diagnostic quiz flows across a Next.js/TypeScript frontend, middleware routing, and backend logging for a nonprofit education platform.
- Developed fully typed TypeScript modules and shipped production features through code review and a Git workflow on a ~70-person cross-functional organization.

## PROJECTS

### WarpServe — GPU Inference Microservice | *CUDA, C++, Python, Docker, Kubernetes*

[Demo](#)

- Hand-wrote a CUDA preprocessing kernel fusing bilinear resize, normalization, and HWC→CHW into one pass — **127× faster** than a scalar CPU reference on a T4, bit-identical; end-to-end only **1.7×**, isolating PCIe transfer as the bottleneck.
- Packaged the FastAPI service as a multi-stage Docker image, keeping the CUDA toolkit out of the shipped artifact: **11.4 GB build stage → 1.52 GB image** (7.5× smaller), running non-root.
- Deployed to Kubernetes with liveness/readiness probes, resource requests, and a horizontal pod autoscaler; scaled **2→8 pods to 88 req/s at 83% scaling efficiency** (p50 907→201 ms) after finding `kubectrl port-forward`, not the pods, capped throughput.

### EdgePulse — Real-Time Forecasting & Anomaly Detection | *C++, Python, ctypes, FastAPI*

[Demo](#)

- Built an ingest → forecast → detect → alert pipeline with a C++ inference core exposed through ctypes, cutting latency ~31% versus Python at ~**508K predictions/sec** (~2 μs/prediction); added median/MAD detection over ring-buffered residuals.

### MarketLab — Vectorized Compute Engine | *Python, pandas, NumPy, pytest*

[Demo](#)

- Replaced a scalar loop with vectorized array operations, achieving ~**30× speedup** (1 ms vs. 30 ms over 50,000 steps) with numerical equivalence verified by pytest.

## TECHNICAL SKILLS

**Languages:** Python, C++, C, CUDA, TypeScript, JavaScript, SQL, MATLAB, Bash, HTML/CSS

**Systems/Tools:** Git/GitHub, Linux/Unix, Docker, Kubernetes, FastAPI, REST APIs, PostgreSQL, Supabase, n8n, AWS, Azure, Google Colab, streamlit, Claude Code

**ML/Data:** PyTorch, NumPy, pandas, scikit-learn, OpenCV, Ultralytics YOLO, ONNX, TensorRT, pytest, ctypes

## PROGRAMS & LEADERSHIP

NVIDIA Summer Bridge (2026) | Jane Street FOCUS (2026) | Goldman Sachs Possibilities Summit | ColorStack Fellow | NSBE Region V | Rice IEEE | Codepath AI110 & TIP 101